GAGNVIST

HVDS AI Gagnathon 2026

Presentation for teams
28.10.2026

Kveikja Glóð EuroHPC Joint Undertaking IHPC National competence center for HPC & AI in Iceland Hagstofa Íslands UNIVERSITY OF ICELAND

Welcome to Gagnaþon 2026

The third Statistics Iceland AI data hackathon

The challenge

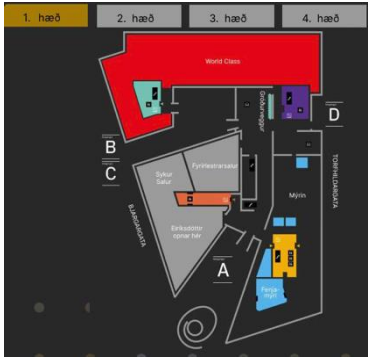
Build a chatbot that makes official statistics easier to find and understand.

The teams

Students, professionals and companies working with open data and AI.

The venue

Gróska · Fenjamýri
Ground-floor south entrance, opposite Eiríksdóttir.



Kveikja · Glóð · EuroHPC Joint Undertaking · IHPC National competence center for HPC & AI in Iceland · Hagstofa Íslands · University of Iceland

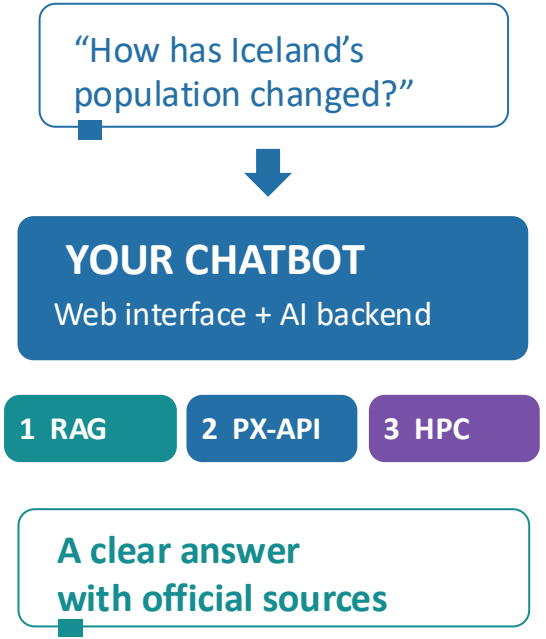
Welcome to the HVDS Gagnaþon

Two days to build a statistical assistant

Turn Statistics Iceland's open data into a chatbot that people can use in everyday language.

Choose prepared RAG, direct PX-API or HPC-optimised RAG. Build a web interface and connect the backend.

Test real questions and present a working demo with clear answers, figures, units and sources.



Kveikja Glóð EuroHPC IHPC National competence center for HPC & AI in Iceland Hagstofa Íslands UNIVERSITY OF ICELAND

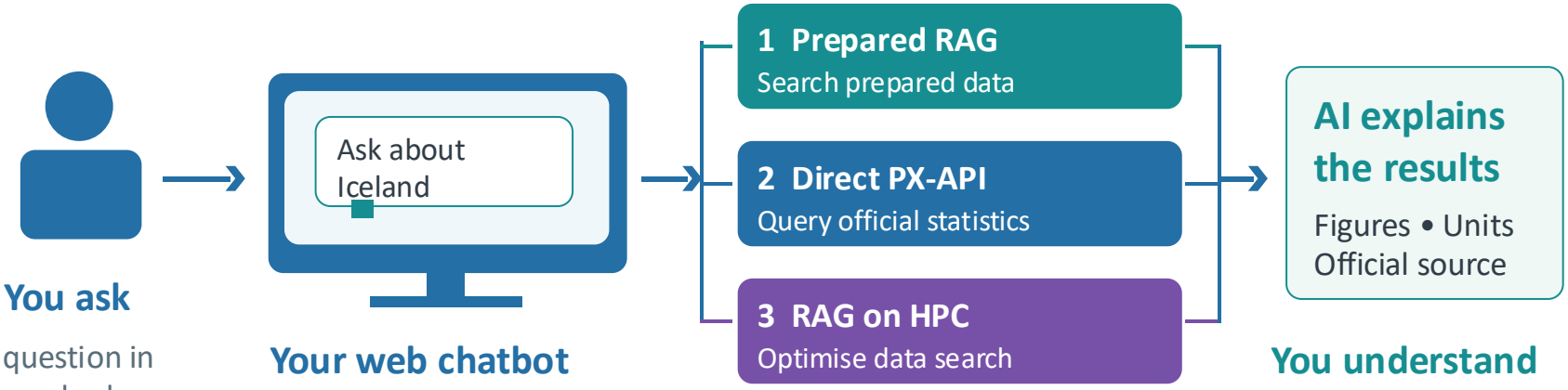
Today’s programme

1 The challenge	The user experience and three project tracks.
2 Getting started	Code, data, README instructions and computing resources.
3 Your demo	Shared deliverables, testing and draft judging criteria.
4 Practical details	Schedule, support, Gagnvist and questions.

From a question to trusted statistics

One chatbot experience. Three ways to find the data.

Population | Prices | Migration



You ask

A question in everyday language

Your web chatbot

You understand

Built on Statistics Iceland's open data • Every track delivers a working prototype

Kveikja | Gljóð | EuroHPC Joint Undertaking | IHPC National competence center for HPC & AI in Iceland | Hagstofa Íslands | University of Iceland

Three tracks, one shared outcome

Choose the route that matches your experience and interests.

TRACK 1

Prepared RAG

Build on a ready-made local retrieval backend.

Focus: interface and integration.

TRACK 2

Direct PX-API

Develop direct access to official statistical tables.

Focus: data selection and backend logic.

TRACK 3

RAG on HPC

Use HPC for embeddings and RAG optimisation.

Focus: retrieval quality and performance.

Track 1 · Prepared RAG

The same chatbot goal, with a different development focus.

Starting point

FastAPI, ChromaDB and SQLite, with a local LLM through Ollama.

FastAPI is a Python framework for building APIs. In your chatbot, it receives web questions, retrieves relevant data, and sends both to Ollama, which runs the AI model locally. FastAPI then returns the generated answer to the user.

Team work

Build a web chatbot, connect the backend and test answers. Optionally extend the data using ingest.py.

Files and setup

GitHub code + README. Unzip hagstofa_db.zip beside app.py: chroma_db/ and data.db.

GitHub link:

Use the model and collection settings specified in the supplied README.

Kveikja Gljóð EuroHPC IHPC National competence center for HPC & AI in Iceland Hagstofa Íslands UNIVERSITY OF ICELAND

Track 2 · Direct PX-API

The same chatbot goal, with a different development focus.

Starting point

A lightweight FastAPI starter that queries Statistics Iceland's PX-API directly.

Team work

Extend table selection and API queries. Build the web chatbot and integrate language-model responses.

Files and setup

GitHub code + README. Fetch data and metadata through PX-API; no local vector database is required.
GitHub link:

Validate table dimensions, filters, units and sources before generating answers.

Track 3 · RAG on HPC

The same chatbot goal, with a different development focus.

Starting point

An additional embedding-based RAG project supplied by the collaborating track provider.

Team work

Use HPC to create or optimise embeddings and retrieval. Connect the results to a web chatbot.

Files and setup

Code, data, HPC access and launch instructions will be provided in the track-specific README.

GitHub link:

Provider details are pending. Compare retrieval quality and performance in your demo.

Getting started · All tracks

Use the README for your track as the setup reference.

Get the materials

Code and instructions on GitHub. Larger data archives on Google Drive, linked from the README.

Prepare your runtime

Tracks 1–2: Python 3.10+, venv and requirements.txt.
Track 1 also needs its databases and Ollama / Llama 3.

Start and test

Tracks 1–2: uvicorn app:app --reload, then /docs or cURL.
Track 3: follow the provider's HPC environment and job instructions.

Final repository links and Track 3 access instructions are still to be supplied.

Show a working solution · Every track

Use the same demonstration structure so the approaches are easy to compare.

1 Ask and answer

Show a real question in your web UI, the retrieved values and an answer with units and an official source.

2 Explain your route

Track 1: local RAG. Track 2: PX-API table selection.
Track 3: embeddings, retrieval and the role of HPC.

3 Show evidence

Check the answer against the source. Show an ambiguous query, response time and a known limitation.

Judging criteria · Draft

Proposed common criteria for all three tracks, each scored from 1 to 5. We expect the submitted solution to be available for use free of charge. You may use any tools of your choice to develop it.

Functionality	Correct retrieval and answers
Presentation	Clear, readable interface
Usability	Easy questions and navigation
Clarity	Understandable answers and sources
Deployment	A reproducible, workable solution
User value	A useful statistical use case

Total: 6–30 points. Assess the result and evidence, not the amount of hardware used.

Final presentations and the winning solution

Shortlisting and presentation format remain a draft.

Submit

A working prototype and 5–7 slides explaining the use case, approach, results and limitations.

Present

Proposed shortlist: five teams. Each selected team has 10 minutes to demonstrate its solution.

The winner

The highest-scoring team receives a prize and presents its solution at Gagnvist on 26 November.

Schedule · 28 October to 1 November

All times are Icelandic local time.

Wed 28 Oct · 16:00	Online introduction for registered teams.
Fri 30 Oct · 15:00	Kickoff and release of project materials and data.
Sat 31 Oct · 08:00	Gróska opens. Saturday access until 22:00.
Sun 1 Nov · 20:00	Team presentations · 10 minutes per team.
Sun 1 Nov · 22:00	Hackathon ends.

Support and Gagnvist

During the event

A team chat group will carry announcements and provide a place for questions and clarifications.

Gagnvist · 26 Nov

The winning solution will be presented at the conference. The exact presentation slot will be confirmed.

Contacts

ingolfuh@hi.is sma41@hi.is
thorarnar@hi.is seh@hi.is

Event information: vidburdir.hagstofa.is/datathon2026

Questions and next steps

Choose your track	Prepared RAG, direct PX-API or RAG on HPC.
Read and prepare	Follow the track README and check the code, data and compute access before you start.
Build and demonstrate	A usable chatbot with clear answers supported by Statistics Iceland’s official data.

Presentation of tracks and Q&A

Presetation of Track 3, Track 2 and Track 1